Certified Lean Six Sigma Yellow Belt (CLSSYB) Training

WSQ Course Code: TGS-2025053922

Conducted by Tertiary Infotech Academy Pte Ltd · UEN: 201200696W

TSC: Quality Process Control · ELE-QUA-5006-1.1

Version v4 · 19 July 2026 · Trainer: Dr. Alfred Ang

 · www.tertiarycourses.com.sg

COURSE ADMINISTRATION

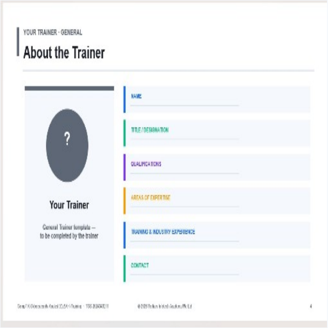
Welcome & Housekeeping

Digital Attendance (Mandatory)

1. Trainer displays the SSG digital attendance QR code
2. Scan the QR code with your phone camera
3. Key in your NRIC/FIN and submit
4. Repeat for AM, PM and the assessment
5. Keep 75% attendance to stay eligible for funding

YOUR TRAINER · GENERAL

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS
PhD; Certified Lean Six Sigma practitioner and trainer.

DELIVERS
WSQ courses on Lean Six Sigma, quality management and data analytics.

EXPERIENCE
Process improvement across manufacturing, service and technology sectors.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Your experience with process improvement or quality work (if any).
- One process at work that frustrates you — we may use it as your course scenario.

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required for certification.

COURSE PORTAL · lms-tms.tertiaryinfotech.com

Download Your Course Material

1. Go to lms-tms.tertiaryinfotech.com
2. Sign in with the account details given in class
3. Open this course from your dashboard
4. Download the slides, Learner Guide and lab files
5. Keep them open — the assessment is open book

SCHEDULE · 9:30am-6:30pm with a 1-hour lunch

Lesson Plan — 2 Days, 8 Hours per Day

Day 1

- Day 1 — Six Sigma foundations, Define phase and process mapping
  - Digital attendance (AM) · Introductions
  - Foundations: Quality, Lean, Six Sigma, belts, DMAIC
  - Lab 1 — Yellow Belt role and scenario
  - DEFINE: VOC, CTQ, charter, SIPOC, process mapping
  - Labs 2-5 — VOC/waste, SIPOC, PDCA charter, DMAIC Define

Day 2

- Day 2 — Measure, Analyse, Improve, Control and assessment
  - Digital attendance (AM)
  - MEASURE: wastes, data types, check sheets, DPMO
  - Lab 6 — Data collection and KPIs
  - ANALYZE: Pareto, run charts, 5 Whys, Fishbone
  - Labs 7-8 — Data analysis and root cause
  - IMPROVE & CONTROL: 5S, poka-yoke, control plan, A3
  - Labs 9-10 · Revision · Final Assessment

WSQ SKILLS FRAMEWORK

Skills Framework — TSC: Quality Process Control

- TSC Code: ELE-QUA-5006-1.1
- A1: Define project to meet process performance.
- A2: Establish project scope of work and the number of hours based on organisational requirements.
- A3: Analyse process performance data to identify root causes of variation.
- A4: Measure process performance against defined quality standards.
- A5: Recommend improvement and control actions to sustain process performance.
- K1: Lean and Six Sigma concepts, wastes, value and variation.
- K2: Quality tools and techniques for process improvement.

WHAT YOU'LL ACHIEVE

Learning Outcomes

1
LO1 — Define & scope
Define a project and establish scope using Define, Measure and Analyse.

2
LO2 — Lean & Six Sigma concepts
Apply value, waste, defects and variation to a work process.

3
LO3 — Map the process
Use SIPOC, process maps and value stream maps to expose handoffs.

4
LO4 — Measure & analyse
Use check sheets, Pareto, run charts and process metrics.

5
LO5 — Root cause
Identify root causes with 5 Whys, Fishbone and evidence.

6
LO6 — Improve & control
Recommend improvement and control actions that sustain the gain.

COURSE ROADMAP

Course Outline — We Follow DMAIC End to End

D → M → A → I → C

Define

- Foundations + Define
- VOC, CTQ, charter
- SIPOC, process maps
- Labs 1-5

Measure

- 8 wastes, data types
- Collection plan
- Yield, DPMO, sigma
- Lab 6

Analyze

- Variation
- Pareto, run charts
- 5 Whys, Fishbone
- Labs 7-8

Improve

- Solution selection
- 5S, poka-yoke
- Standard work
- Lab 9

Control

- Control plan
- Visual management
- A3 and handover
- Lab 10

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 60 minutes, open book.
- Case Study (CS) — applied Lean Six Sigma tasks, 90 minutes, open book.
- Format: Open Book — slides, Learner Guide and approved materials only.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (SAQ), then the Case Study — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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FOUNDATIONS

Six Sigma Foundations

Quality · Lean · Six Sigma · Lean Six Sigma · Belt roles · The DMAIC roadmap

FOUNDATIONS · QUALITY

What is Quality?

Before we can improve quality, we have to agree what the word actually means.

WHAT IS QUALITY?

Three Common Answers — and Why Only One Holds Up

Defect-free output

"Quality means zero defects"

- Necessary, but not sufficient.
- A product can be flawless and still not sell.
- Defect-free against WHOSE standard?

Meeting set standards

"Quality means meeting the spec"

- Better — it is measurable.
- But a spec can be set at the wrong level.
- Meeting a spec nobody wants is waste.

Meeting customer expectations

"Quality means the customer is satisfied"

- This is the Six Sigma definition.
- The customer sets the standard.
- Everything else follows from this.

FOUR KEY ACTIVITIES

Quality in a Nutshell

1

Understand requirements

Find out what the customer actually needs — in their own words.

2

Design to satisfy them

Build the product or service so it meets those requirements.

3

Deliver consistently

Reduce variation so every customer gets the same experience.

4

Improve continuously

Requirements move; the process has to keep up.

FOUNDATIONS · LEAN

What is Lean?

Lean maximises customer value by systematically removing everything the customer would not pay for.

WHAT IS LEAN?

Lean — The Core Idea

1 **Origin**
The Toyota Production System — refined in Japanese manufacturing from the 1950s.

2 **Focus**
Speed and flow: shorten the time from customer request to delivery.

3 **Enemy**
Waste (muda) — any effort that consumes resources but creates no customer value.

4 **Method**
See the process, find the waste, remove it, then standardise the improvement.

5 **Applies to**
Manufacturing AND service — hospitals, banks, IT service desks, government.

6 **Yellow Belt use**
Waste walks, process mapping, 5S and small Kaizen improvements.

The Five Lean Principles

1
Specify VALUE from the customer's point of view

2
Map the VALUE STREAM and expose the waste in it

3
Create FLOW so work moves without interruption

4
Let the customer PULL work rather than pushing it

5
Pursue PERFECTION through continuous improvement

FOUNDATIONS · SIX SIGMA

What is Six Sigma?

A disciplined, data-driven method to reduce variation and defects — so the process delivers the same result every time.

WHAT IS SIX SIGMA?

Six Sigma — The Core Idea

1

Origin

Motorola, 1986 — later made famous by General Electric under Jack Welch.

2

Focus

Consistency: reduce the variation that creates defects.

3

Enemy

Variation — the spread that turns a good average into an unreliable experience.

4

Method

DMAIC, driven by data and statistics rather than opinion.

5

The target

No more than 3.4 defects per million opportunities (DPMO).

6

Yellow Belt use

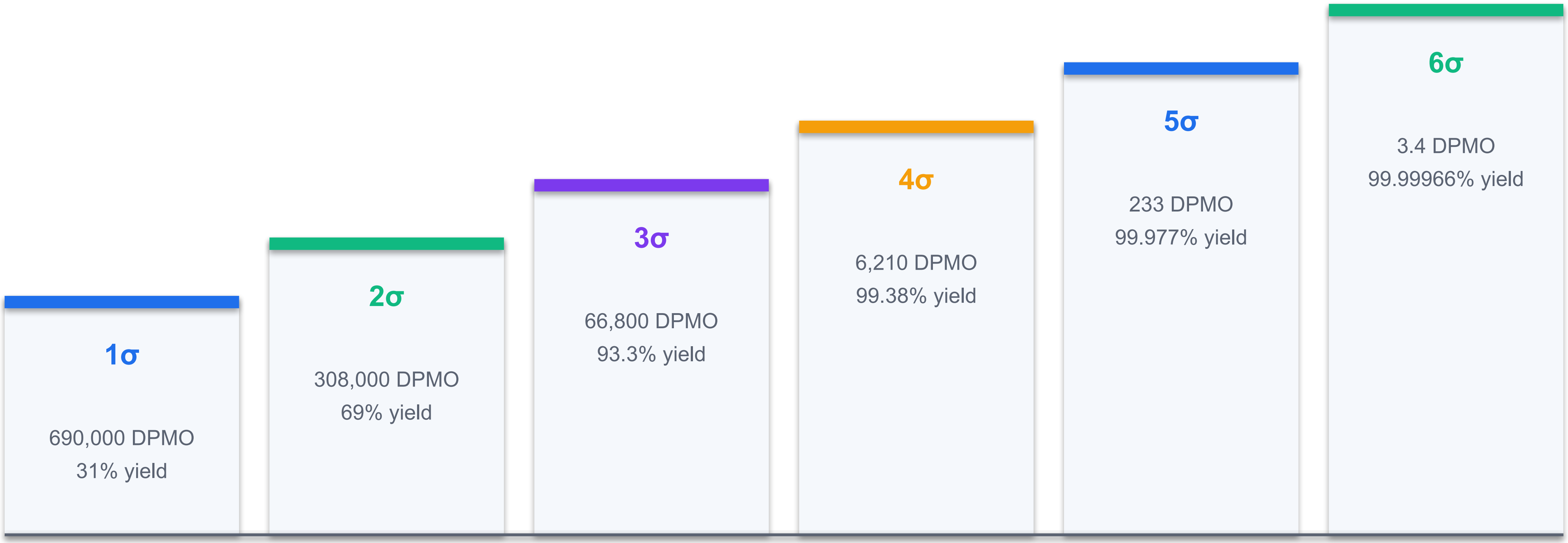
Data collection, Pareto and run charts, root cause analysis.

THE STATISTICAL MEANING

Six Sigma as Standard Deviation

Sigma (σ) measures spread. The more sigmas that fit between the process mean and the specification limits, the less likely the process is to produce a defect.

Sigma Level vs Defects per Million Opportunities



Each sigma level is a step change, not an increment — moving 3σ → 4σ removes about 90% of the defects.

THE COST OF 'GOOD ENOUGH'

Why 99% Is Not Good Enough

99% quality

- At 99% quality (about 3.8σ)
  - 20,000 lost articles of mail per hour
  - Unsafe drinking water almost 15 minutes a day
  - 5,000 incorrect surgical operations per week
  - Two short or long landings at major airports daily

99.99966% quality

- At Six Sigma quality (3.4 DPMO)
  - 7 lost articles of mail per hour
  - Unsafe drinking water 1 minute every 7 months
  - 1.7 incorrect surgical operations per week
  - One short or long landing every 5 years

THE COMBINED METHOD

Lean vs Six Sigma vs Lean Six Sigma

Lean

- Removes waste
- Improves speed and flow
- Question: is this step worth doing?
- Tools: VSM, 5S, Kaizen, poka-yoke

+

Six Sigma

- Reduces variation
- Improves consistency and accuracy
- Question: why does the result vary?
- Tools: DMAIC, Pareto, SPC, root cause

▼

Lean Six Sigma

Faster AND more consistent — remove the waste, then control the variation that remains.

HISTORY

A Short History of Lean Six Sigma

1950s
Toyota Production System — Lean is born

1986
Motorola coins Six Sigma

1995
GE adopts Six Sigma company-wide

2000s
Lean and Six Sigma merge

Today
Applied across service, healthcare, finance and IT

THE CENTRAL EQUATION

$Y = f(X)$

The output Y is a function of the process inputs X. To change the result, change the inputs that drive it — not the result itself.

OUR COURSE SCENARIO

Y = f(X) in Practice — The Contoso Service Desk

1

Y — the output

Ticket assignment time, measured in hours from submission to assignment.

2

X — triage rules

How clearly the routing rules are written and understood.

3

X — staffing

How many agents are available at peak submission times.

4

X — ticket quality

How complete the information is when the ticket arrives.

WHO DOES WHAT

The Lean Six Sigma Belt Pathway

White Belt
Awareness of Six Sigma; supports local problem solving

Yellow Belt
Knows the basic tools; supports projects and runs small improvements

Green Belt
Leads smaller projects; assists Black Belts with analysis

Black Belt
Leads complex projects full time; coaches Green and Yellow Belts

Master Black Belt
Trains and mentors Belts; owns deployment strategy

This course certifies you at Yellow Belt — the level that supports projects and runs small improvements.

LSS YELLOW BELT

Your Role — the Lean Six Sigma Yellow Belt

1 **Has basic knowledge**
Understands the DMAIC roadmap and the 7 basic quality tools.

2 **Supports the project**
Contributes process knowledge as the subject-matter expert.

3 **Collects the data**
Runs check sheets and data collection under the project leader.

4 **Maps the process**
Builds SIPOC and process maps that expose handoffs and waste.

5 **Runs small improvements**
Leads Kaizen or PDCA improvements within their own work area.

6 **Does not lead DMAIC**
Larger, cross-functional DMAIC projects belong to a Green or Black Belt.

THE COURSE ROADMAP · WE FOLLOW THIS ORDER

The DMAIC Roadmap

D → M → A → I → C

Define

- Define the problem
- Capture VOC and CTQ
- Charter and scope
- Map with SIPOC

Measure

- Plan data collection
- Measure the baseline
- Calculate yield/DPMO
- Find the waste

Analyze

- Analyse the data
- Pareto and run charts
- 5 Whys and Fishbone
- Prove the root cause

Improve

- Generate solutions
- Select and pilot
- 5S and poka-yoke
- Standardise the work

Control

- Build the control plan
- Visual management
- Hand over to the owner
- Sustain the gain

DMAIC vs PDCA — Which One Do I Use?

Small improvement

- PDCA — Plan, Do, Check, Act
  - For small, local, fast improvements
  - Days to a few weeks
  - A Yellow Belt can run this
  - Lower data requirement
  - Used in Lab 4

Full project

- DMAIC — the full roadmap
  - For larger, cross-functional problems
  - Weeks to months
  - Led by a Green or Black Belt
  - Data and statistics required
  - Supported by you from Lab 5 onward

PDCA CYCLE

PDCA — The Small Improvement Cycle

1

PLAN — define the problem and plan the change

2

DO — run the change at small scale

3

CHECK — measure whether it actually worked

4

ACT — adopt, adapt or abandon, then repeat

PROJECT SELECTION

What Makes a Good Yellow Belt Project?

1. **Related to your daily work**
   You know the process and can observe it directly.

2. **Manageable in timeframe**
   Can be completed in weeks, not months.

3. **Aligned to business goals**
   Someone in management cares about the result.

4. **Data is available**
   You can measure the baseline and the improvement.

FOUNDATIONS · HANDS-ON

Yellow Belt Role, Certification Paths, and Improvement Scenario

LAB 1

Establish what a Lean Six Sigma Yellow Belt is accountable for, how the belt pathway progresses from White to Master Black Belt, and choose a small, manageable improvement scenario to carry through the whole course.

You'll build

A Yellow Belt responsibility table and a selected improvement scenario.

Tools & techniques: Belt role matrix, CSSC certification pathway, project selection criteria

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 1 OF 5

1

Define the Yellow Belt role — list at least four responsibilities and your contribution to each.

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 2 OF 5

2

Compare the White, Yellow, Green, Black and Master Black Belt pathways in a table.

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 3 OF 5

3

Review the Contoso Service Desk scenario: tickets are slow to be assigned and status is unclear.

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 4 OF 5

4

Test your scenario against the 'good project' criteria — day-to-day work, manageable, aligned to business goals, data available.

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 5 OF 5

5

Record why a Yellow Belt supports rather than leads this improvement.

LAB 1 · VERIFY

Yellow Belt Role, Certification Paths, and Improvement Scenario

✅ Check your work

You can state the Yellow Belt role in one sentence and justify your scenario against all four selection criteria.

D

DMAIC · DEFINE

Define — Scope the Problem

VOC · CTQ · Project charter · Problem statement · Scope · SIPOC · Process mapping

DEFINE · 25% OF THE COURSE

Key Concepts — Define

1 **Voice of the Customer**
Capture customer needs in their own words before deciding anything.

2 **Critical to Quality**
Translate each VOC need into a specific, measurable CTQ requirement.

3 **Project charter**
Problem statement, goal, scope, team and benefit — the project's contract.

4 **Problem statement**
Process, time period, measurable issue and impact — and never a solution.

5 **SIPOC**
The macro as-is map: Suppliers, Inputs, Process, Outputs, Customers.

6 **Process mapping**
Flowcharts and swimlanes expose the handoffs where delay and defects are born.

DMAIC · D — DEFINE

Define — What problem are we solving, and for whom?

The Define phase converts a vague complaint into a scoped, measurable problem statement everyone agrees on.

Steps in the Define Phase

1. Capture the Voice of the Customer
2. Translate VOC into CTQ requirements
3. Write the problem and goal statements
4. Charter and scope the project
5. Map the process with SIPOC

DEFINE · VOC

Voice of the Customer (VOC)

The needs and expectations of the customer, expressed in the customer's own language.

SOURCES OF VOC

Where VOC Comes From

1

Surveys

Structured and at scale, but limited to the questions you thought to ask.

2

Interviews

Rich and open-ended; best for understanding the 'why' behind a complaint.

3

Complaints & tickets

Already in your system — the cheapest and most honest VOC source.

4

Direct observation

Go and watch the work happen; customers cannot always articulate the problem.

VOC → Need → CTQ: The Translation

Voice of the Customer

- The customer says (VOC)
  - "I don't want to wait too long"
  - "I don't want cold pizza"
  - "I never know where my ticket is"
  - Vague, emotional, unmeasurable

Critical to Quality

- The measurable requirement (CTQ)
  - Delivery in 30 minutes or less
  - Pizza at 32°C minimum on arrival
  - Status update within 4 working hours
  - Specific, numeric, testable

CTQ QUALITY CHECK

What Makes a Good CTQ?

1. **Specific**
   Names exactly one characteristic of the output.

2. **Measurable**
   Has a number and a unit you can actually record.

3. **Linked to VOC**
   Traces back to something a customer genuinely said.

4. **Has a target and limit**
   States the target value and the acceptable range around it.

The Project Charter

1. **Business case**
   Why this project matters and what it is worth.

2. **Problem statement**
   What is wrong, quantified, over a stated time period.

3. **Goal statement**
   Metric, baseline, target and date.

4. **Scope**
   What is in, and explicitly what is out.

5. **Team and roles**
   Sponsor, project leader, Yellow Belt support and SMEs.

6. **Milestones**
   The dates each DMAIC phase is expected to complete.

Problem Statements — Good vs Bad

Avoid this

- Weak problem statement
  - "The service desk is too slow."
  - No process named
  - No time period
  - No measure or baseline
  - No customer impact stated
  - Hints at a solution ("we need more staff")

Aim for this

- Strong problem statement
  - "Between 1 Jan and 31 Mar, tickets took an average of 9.4 working hours to be assigned against a 4-hour target, affecting 38% of employees and generating 210 chase-up contacts."
  - Process, period, measure, baseline and impact — and no solution.

PROBLEM STATEMENT ANATOMY

The Four Components of a Problem Statement

1 **What**
The specific process and the defect or gap observed.

2 **When**
The time period the data covers.

3 **How much**
The measured baseline against the requirement.

4 **So what**
The impact on the customer and the business.

DEFINE · SIPOC

SIPOC — the macro 'as-is' map

One page that shows who supplies the process, what goes in, what happens, what comes out, and who receives it.

DEFINE · WORKED EXAMPLE

SIPOC — Contoso Service Desk Ticket Assignment

S — Suppliers

- Employees
- Line managers
- HR onboarding
- Monitoring tools

I — Inputs

- Ticket form
- Category
- Priority
- Asset details
- Contact info

P — Process

- 1. Ticket submitted
- 2. Auto-categorised
- 3. Triage review
- 4. Assigned to agent
- 5. Agent acknowledges

O — Outputs

- Assigned ticket
- Assignment notification
- Priority classification
- SLA clock started

C — Customers

- Employee
- Support agent
- Service desk manager
- IT leadership

Read left to right: suppliers provide the inputs, the process transforms them, and the outputs go to the customers.

SIPOC · METHOD

How to Build a SIPOC

1. Name the process and agree its start and stop points
2. List the high-level steps — 5 to 7, no more
3. Identify the outputs and who receives them
4. Identify the inputs each step needs
5. Identify who supplies each input

Why Process Maps Matter

1

Shows the real process

Not the process in the procedure manual — the one people actually follow.

2

Exposes handoffs

Delay and defects are usually created where work changes hands.

3

Builds shared understanding

The team often discovers they each believed a different process.

4

Finds the waste

Rework loops, waiting and duplicated approvals become visible.

5

Sets the baseline

You cannot improve a process you have not described.

6

Guides data collection

The map tells you where to put your measurement points.

CHOOSING THE RIGHT MAP

Three Levels of Process Map

SIPOC

Macro / 30,000 feet

- 5-7 steps on one page.
- Sets scope and boundaries.
- Built first, in the Define phase.

Process flowchart

Detailed / ground level

- Every step, decision and loop.
- Uses standard symbols.
- Reveals rework and delay.

Swimlane map

Detailed + ownership

- One lane per actor or team.
- Handoffs cross the lanes.
- Best for finding delay in handoffs.

READING A PROCESS MAP

Standard Process Map Symbols

1. **Oval — Terminator**
   The start or the end of the process.

2. **Rectangle — Process step**
   An activity or task being performed.

3. **Diamond — Decision**
   A yes/no branch point in the flow.

4. **Arrow — Flow**
   The direction work moves between steps.

5. **Parallelogram — Data**
   An input to, or output from, a step.

6. **D shape — Delay**
   Work waiting in a queue — usually where the waste is.

WHAT YOU'LL DO

Hands-On Labs — Define

1. **Lab 2 — Lean, Six Sigma, Waste, Voice of Customer, and**
   A VOC-to-CTQ translation table, a value-added analysis, and a waste wa

2. **Lab 3 — SIPOC, Process Mapping, Handoffs, and SME Supp**
   A completed SIPOC and a detailed process map with pain points marked.

3. **Lab 5 — DMAIC Overview, Problem Statement, Scope, and**
   A DMAIC phase table, refined problem statement, stakeholder map and be

4. **Lab 4 (elective) — PDCA Small Improvement Project Charter**
   A one-page PDCA charter with problem statement, goal, scope and succes

5. **Lab 11 (elective) — Affinity Diagram and Kano Analysis**
   An Affinity Diagram of clustered VOC themes and a Kano classification

Lean, Six Sigma, Waste, Voice of Customer, and Value

LAB 2

Separate what the customer actually values from the waste that surrounds it. Capture the Voice of the Customer, translate it into measurable CTQ requirements, and run a waste walk using the eight wastes (DOWNTIME).

You'll build

A VOC-to-CTQ translation table, a value-added analysis, and a waste walk log.

Tools & techniques: VOC, CTQ tree, DOWNTIME eight wastes, value-added analysis

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 1 OF 6

1

Capture the Voice of the Customer — record at least five verbatim customer statements.

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 2 OF 6

2

Translate each VOC statement into a need, then into a measurable CTQ requirement.

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 3 OF 6

3

Classify each process activity as value-added, business-value-added or non-value-added.

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 4 OF 6

4

Conduct a waste walk and tag each observation against the eight wastes (DOWNTIME).

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 5 OF 6

5

Distinguish a defect (output fails CTQ) from waste (effort the customer will not pay for).

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 6 OF 6

6

Identify which single waste type appears most often in your scenario.

LAB 2 · VERIFY

Lean, Six Sigma, Waste, Voice of Customer, and Value

✅ Check your work

Every CTQ is measurable with a target, and each waste observation is tagged to one of the eight waste types.

DMAIC · DEFINE · HANDS-ON

SIPOC, Process Mapping, Handoffs, and SME Support

LAB 3

Build the macro 'as-is' view with SIPOC, then drill into a detailed process map that shows every actor, system and handoff — the points where delay and defects are actually created.

You'll build

A completed SIPOC and a detailed process map with pain points marked.

Tools & techniques: SIPOC, process flowchart, swimlane map, standard process symbols

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 1 OF 6

1

Set the process boundaries — agree the explicit start and stop points first.

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 2 OF 6

2

Build the SIPOC: Suppliers, Inputs, Process (5-7 high-level steps), Outputs, Customers.

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 3 OF 6

3

Expand into a detailed process map: Step, Actor, Activity, System, Handoff.

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 4 OF 6

4

Draw the same flow as a swimlane map so each actor's lane makes handoffs obvious.

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 5 OF 6

5

Mark pain points — delays, rework loops, unclear ownership and handoff risks.

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 6 OF 6

6

Prepare SME notes for the Green Belt: what you observed and what needs validation.

LAB 3 · VERIFY

SIPOC, Process Mapping, Handoffs, and SME Support

✅ Check your work

Your SIPOC has all five columns populated and every handoff on the detailed map has a named owner on both sides.

Elective — PDCA Small Improvement Project Charter

LAB 4

ELECTIVE

Not every improvement needs a full DMAIC project. Use PDCA to charter a small, fast improvement that a Yellow Belt can run under guidance — with a problem statement, a measurable goal and a defined check point.

You'll build

A one-page PDCA charter with problem statement, goal, scope and success measure.

Tools & techniques: PDCA cycle, problem/goal statements, scope in-out, stakeholder list

Elective — PDCA Small Improvement Project Charter

STEP 1 OF 6

1

Map the four PDCA phases to concrete actions for your scenario.

Elective — PDCA Small Improvement Project Charter

STEP 2 OF 6

2

Write the problem statement: process, time period, measurable issue and impact — no solution.

Elective — PDCA Small Improvement Project Charter

STEP 3 OF 6

3

Write a measurable goal statement: metric, baseline, target and date.

Elective — PDCA Small Improvement Project Charter

STEP 4 OF 6

4

Define scope with an explicit in-scope / out-of-scope table to prevent scope creep.

Elective — PDCA Small Improvement Project Charter

STEP 5 OF 6

5

Identify stakeholders and the decision you will make after the Check phase.

Elective — PDCA Small Improvement Project Charter

STEP 6 OF 6

6

Confirm the improvement is small enough to test within two weeks.

Elective — PDCA Small Improvement Project Charter

✅ Check your work

Your goal statement contains a metric, a baseline, a target and a date, and your problem statement names no solution.

DMAIC Overview, Problem Statement, Scope, and Stakeholders

LAB 5

Step up from PDCA to the full DMAIC roadmap. Clarify what each phase delivers, where a Yellow Belt contributes most, and prepare defensible Define-phase information for the project leader.

You'll build

A DMAIC phase table, refined problem statement, stakeholder map and benefit estimate.

Tools & techniques: DMAIC roadmap, project charter, stakeholder analysis, CTQ linkage

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 1 OF 6

1

Summarise all five DMAIC phases: purpose, key deliverable and Yellow Belt support role.

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 2 OF 6

2

Refine the problem statement so it is specific, measurable and solution-free.

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 3 OF 6

3

Identify stakeholders and classify each by influence and interest.

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 4 OF 6

4

Define the project scope and state the expected benefit in business terms.

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 5 OF 6

5

Link the problem back to the CTQ requirements captured in Lab 2.

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 6 OF 6

6

Confirm which DMAIC phases a Yellow Belt can support most strongly.

LAB 5 · VERIFY

DMAIC Overview, Problem Statement, Scope, and Stakeholders

✅ Check your work

Your problem statement passes the 'no solution named' test and every stakeholder has a defined engagement approach.

DMAIC · DEFINE · HANDS-ON

Elective — Affinity Diagram and Kano Analysis

LAB 11

ELECTIVE

Two Define-phase tools from the original course. Use an Affinity Diagram to cluster unstructured VOC input into themes, then apply Kano Analysis to classify requirements as Must-Be, One-Dimensional or Delighter.

You'll build

An Affinity Diagram of clustered VOC themes and a Kano classification table.

Tools & techniques: Affinity Diagram, Kano Analysis, VOC clustering

Elective — Affinity Diagram and Kano Analysis

STEP 1 OF 4

1

Write each VOC statement on a separate note — one idea per note.

Elective — Affinity Diagram and Kano Analysis

STEP 2 OF 4

2

Cluster related notes into natural groups without talking, then name each group.

Elective — Affinity Diagram and Kano Analysis

STEP 3 OF 4

3

Classify each requirement as Must-Be, One-Dimensional or Delighter.

Elective — Affinity Diagram and Kano Analysis

STEP 4 OF 4

4

Plot the requirements on the Kano diagram and identify where to invest first.

LAB 11 · VERIFY

Elective — Affinity Diagram and Kano Analysis

✅ Check your work

Every VOC note sits in exactly one named affinity group and carries a Kano classification.

PHASE RECAP

Recap — Define

- Voice of the Customer — Capture customer needs in their own words before deciding anything.
- Critical to Quality — Translate each VOC need into a specific, measurable CTQ requirement.
- Project charter — Problem statement, goal, scope, team and benefit — the project's contract.
- Problem statement — Process, time period, measurable issue and impact — and never a solution.
- SIPOC — The macro as-is map: Suppliers, Inputs, Process, Outputs, Customers.
- Process mapping — Flowcharts and swimlanes expose the handoffs where delay and defects are born.

M

DMAIC · MEASURE

Measure — Quantify Performance

The 8 wastes · Data types · Data collection plan · Check sheets · Yield · DPMO · Sigma level

Key Concepts — Measure

1. **The 8 wastes**
DOWNTIME — Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

2. **Types of data**
Continuous vs discrete; nominal vs ordinal — the data type drives the tool choice.

3. **Data collection plan**
What, who, when and how — with operational definitions to remove ambiguity.

4. **Check sheets**
A simple structured form is the most reliable way to capture process data.

5. **Process metrics**
Yield, DPU, DPO and DPMO quantify how well the process actually performs.

6. **Sigma level**
Convert DPMO into a sigma level to benchmark the process against Six Sigma.

DMAIC · M — MEASURE

Measure — How big is the problem, really?

The Measure phase replaces opinion with a trustworthy baseline you can improve against.

Steps in the Measure Phase

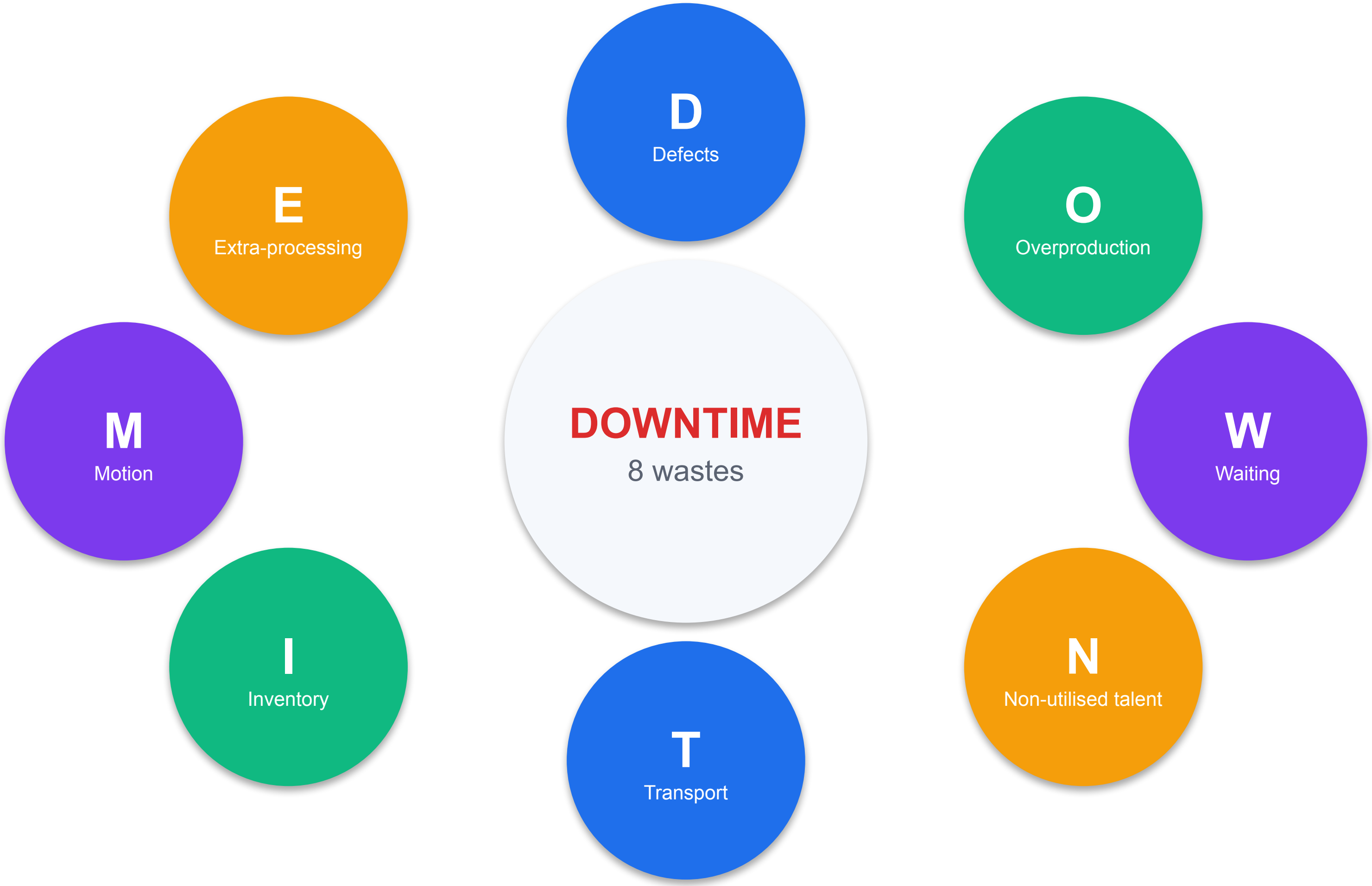
1. Identify the waste in the current process
2. Decide what to measure and define it precisely
3. Build the data collection plan and check sheet
4. Collect the baseline data
5. Calculate yield, DPMO and sigma level

MEASURE · WASTE (MUDA)

What is Waste?

Anything beyond the minimum information, equipment, material and effort absolutely required to add value for the customer.

The Eight Wastes of Lean — DOWNTIME



Waste is any effort the customer would not willingly pay for.

DOWNTIME IN PRACTICE

The Eight Wastes in a Service Process

1 **D — Defects**
Wrong ticket category, incomplete information, work that must be redone.

2 **O — Overproduction**
Reports nobody reads; producing output before it is needed.

3 **W — Waiting**
Tickets sitting in a queue waiting for triage or approval.

4 **N — Non-utilised talent**
Skilled agents doing routine data entry a form could handle.

5 **T — Transport**
Tickets bouncing between teams before reaching the right owner.

6 **I — Inventory**
A growing backlog of unassigned tickets.

7 **M — Motion**
Switching between four systems to resolve one ticket.

8 **E — Extra-processing**
Approvals and checks that add no value to the customer.

Value-Added vs Non-Value-Added

Value-added (VA)

The customer would pay for it

- Changes the product or service.
- Done right the first time.
- The customer cares that it happened.
- Example: diagnosing the actual fault.

Business-value-added (BVA)

Required, but not by the customer

- Needed for legal or regulatory reasons.
- Required to run the business.
- Minimise it — you cannot remove it.
- Example: mandatory audit logging.

Non-value-added (NVA)

Pure waste — remove it

- Consumes time and resource for nothing.
- The customer would never pay for it.
- Attack this first.
- Example: a ticket waiting in a queue.

AN IMPORTANT DISTINCTION

Waste is not the same as a defect.

A defect is an output that fails the CTQ requirement. Waste is any effort along the way that the customer would not willingly pay for. Every defect creates waste — but most waste is not a defect.

Types of Data — and Why It Matters

Continuous data

Measured on a scale

- Any value within a range.
- Time, cost, temperature, length.
- More information per data point.
- Needs a smaller sample size.

Discrete / attribute data

Counted in whole units

- Counts and categories only.
- Number of defects, pass/fail.
- Less information per data point.
- Needs a much larger sample.

THE FOUR MEASUREMENT SCALES

Nominal, Ordinal, Interval, Ratio

1

Nominal

Labels with no order — ticket category, agent name, department.

2

Ordinal

Ordered categories, but the gaps are not equal — priority P1/P2/P3, satisfaction 1-5.

3

Interval

Equal gaps, but no true zero — temperature in °C, calendar dates.

4

Ratio

Equal gaps and a true zero — assignment time, cost, number of tickets.

WHY THIS MATTERS

The data type decides the tool.

Choose your chart, your test and your sample size AFTER you know what kind of data you are holding — not before.

MEASURE · PLANNING THE DATA

The Data Collection Plan

1 **What**
Which metric, and what exactly counts as one observation.

2 **Why**
Which CTQ or problem statement this metric supports.

3 **How**
Automated report preferred; manual capture only when unavoidable.

4 **Who**
Someone who knows the process and is available to do it consistently.

5 **When**
The frequency and the exact period the data covers.

6 **How many**
The sample size — enough to be representative, not so many it is unaffordable.

MEASURE · OPERATIONAL DEFINITIONS

Operational definitions remove the argument.

"Assignment time" means nothing until you state it precisely: the elapsed working hours from ticket submission timestamp to the first agent assignment timestamp, excluding weekends and public holidays.

Check Sheets — the Simplest Reliable Tool

1. **One row per observation**
   Never summarise as you collect — record the raw event.

2. **Pre-printed categories**
   Tick a box rather than writing free text; free text cannot be counted.

3. **Mutually exclusive**
   Every observation must fall into exactly one category.

4. **Include the context**
   Date, time, shift and who recorded it — you will need this later.

Sampling Techniques

1

Simple random

Every unit has an equal chance of being selected.

2

Stratified random

Split the population into groups, then sample randomly within each group.

3

Systematic

Take every nth unit — simple, but beware of hidden cycles in the data.

4

Cluster

Sample whole naturally occurring groups when individual sampling is impractical.

Process Performance Metrics

Yield	**Yield = (Good units / Total units) × 100**	950 of 1,000 tickets assigned on time → 95% yield
DPU	**DPU = Defects / Units**	600 defects across 1,000 tickets → DPU = 0.6
DPO	**DPO = Defects / (Units × Opportunities)**	600 defects, 1,000 tickets, 3 opportunities each → DPO = 0.2
DPMO	**DPMO = DPO × 1,000,000**	DPO of 0.2 → 200,000 DPMO

An 'opportunity' is any single chance for a defect to occur on one unit — agree the count before you measure.

Yield — Three Different Questions

1. **Classic yield**
   What fraction of units passed final inspection? Ignores any rework along the way.

2. **First pass yield (FPY)**
   What fraction passed with NO rework at any step? The honest number.

3. **Rolled throughput yield (RTY)**
   Multiply the FPY of every step — the probability a unit passes cleanly through the entire process.

4. **Why RTY matters**
   Five steps at 95% each give an RTY of only 77% — small defect rates compound fast.

From DPMO to Sigma Level

Sigma level	**Look DPMO up in the sigma conversion table**	200,000 DPMO ≈ 2.3σ
Our example	**Yield 99.5% → 5,000 DPMO**	5,000 DPMO ≈ 4.1σ
The Six Sigma target	**3.4 DPMO**	99.99966% yield

Sigma level is the common yardstick that lets you compare performance across completely different processes.

Cost of Poor Quality (COPQ)

1

Internal failure

Defects found before the customer sees them — rework, scrap, re-triage.

2

External failure

Defects the customer finds — complaints, escalations, lost trust.

3

Appraisal cost

The cost of inspecting and checking for defects.

4

Prevention cost

The cost of stopping defects happening — training, poka-yoke, better design.

WHAT YOU'LL DO

Hands-On Labs — Measure

1

Lab 6 — Data Collection, KPIs, Check Sheets, and Basic

A data collection plan, an operational definition set and a working ch

2

Lab 12 (elective) — Value Stream Map and Takt Time

A value stream map with lead time, process time and a calculated takt

DMAIC · MEASURE · HANDS-ON

Data Collection, KPIs, Check Sheets, and Basic Metrics

LAB 6

Replace anecdote with evidence. Define KPIs with unambiguous operational definitions, design a check sheet, plan the sampling approach, and understand which data type you are collecting — because the data type drives the tool.

You'll build

A data collection plan, an operational definition set and a working check sheet.

Tools & techniques: KPI definition, operational definitions, check sheets, sampling, data types

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 1 OF 6

1

Classify your data: continuous vs discrete, nominal vs ordinal — and why it matters.

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 2 OF 6

2

Define KPIs with operational definitions, data source and collection frequency.

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 3 OF 6

3

Design a check sheet capturing date, ticket ID, type, assignment time and defect category.

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 4 OF 6

4

Define mutually exclusive defect categories so every observation lands in exactly one.

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 5 OF 6

5

Plan sampling: how many, how often, by whom — and identify possible bias.

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 6 OF 6

6

State which KPI best reflects the customer pain point from your VOC work.

Data Collection, KPIs, Check Sheets, and Basic Metrics

✅ Check your work

Two different people reading your operational definition would record the same value for the same event.

DMAIC · MEASURE · HANDS-ON

Elective — Value Stream Map and Takt Time

LAB 12

ELECTIVE

Extend process mapping into Lean metrics. Map the value stream with process and wait times, calculate process cycle efficiency, then compute takt time to see whether the process can meet customer demand.

You'll build

A value stream map with lead time, process time and a calculated takt time.

Tools & techniques: Value stream mapping, lead time, WIP, takt time, cycle efficiency

Elective — Value Stream Map and Takt Time

STEP 1 OF 4

1

Map the value stream: each step with its process time and the wait time between steps.

Elective — Value Stream Map and Takt Time

STEP 2 OF 4

2

Total the value-added time and the lead time, then compute process cycle efficiency.

Elective — Value Stream Map and Takt Time

STEP 3 OF 4

3

Calculate takt time = available working time / customer demand.

Elective — Value Stream Map and Takt Time

STEP 4 OF 4

4

Compare cycle time against takt time to identify the bottleneck step.

LAB 12 · VERIFY

Elective — Value Stream Map and Takt Time

✅ Check your work

Your lead time equals the sum of all process and wait times, and takt time is expressed per unit.

PHASE RECAP

Recap — Measure

- The 8 wastes — DOWNTIME — Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.
- Types of data — Continuous vs discrete; nominal vs ordinal — the data type drives the tool choice.
- Data collection plan — What, who, when and how — with operational definitions to remove ambiguity.
- Check sheets — A simple structured form is the most reliable way to capture process data.
- Process metrics — Yield, DPU, DPO and DPMO quantify how well the process actually performs.
- Sigma level — Convert DPMO into a sigma level to benchmark the process against Six Sigma.

A

DMAIC · ANALYZE

Analyze — Find the Root Cause

Variation · Pareto · Run charts · 5 Whys · Fishbone · Multi-voting · Evidence

Key Concepts — Analyze

1 **Variation**
Common cause is built into the process; special cause is an external, assignable signal.

2 **Pareto analysis**
The 80/20 rule separates the vital few causes from the trivial many.

3 **Run charts**
Plot the metric over time to reveal trend, shift, cluster and oscillation.

4 **5 Whys**
Ask why repeatedly to drill from the symptom down to an actionable cause.

5 **Fishbone diagram**
Organise candidate causes by category — Manpower, Method, Machine, Material, Measurement.

6 **Evidence testing**
A cause is only a root cause when the data you collected supports it.

DMAIC · A — ANALYZE

Analyze — Why is this happening?

The Analyze phase moves the team from a measured symptom to a proven, actionable root cause.

Steps in the Analyze Phase

1. Study the data and the process map
2. Generate a list of potential causes (Xs)
3. Organise the causes with a Fishbone diagram
4. Drill down with the 5 Whys
5. Test each candidate cause against the evidence

Two Kinds of Variation — and Two Different Responses

Common cause

Built into the process

- Always present, random, predictable range.
- The process is stable but imperfect.
- Response: change the PROCESS.
- Reacting to individual points makes it worse.

Special cause

An external, assignable signal

- Unusual, traceable to a specific event.
- Something changed that should not have.
- Response: investigate THAT event.
- Fix the cause, then remove it for good.

THE MOST COMMON ANALYSIS MISTAKE

Do not react to a single data point.

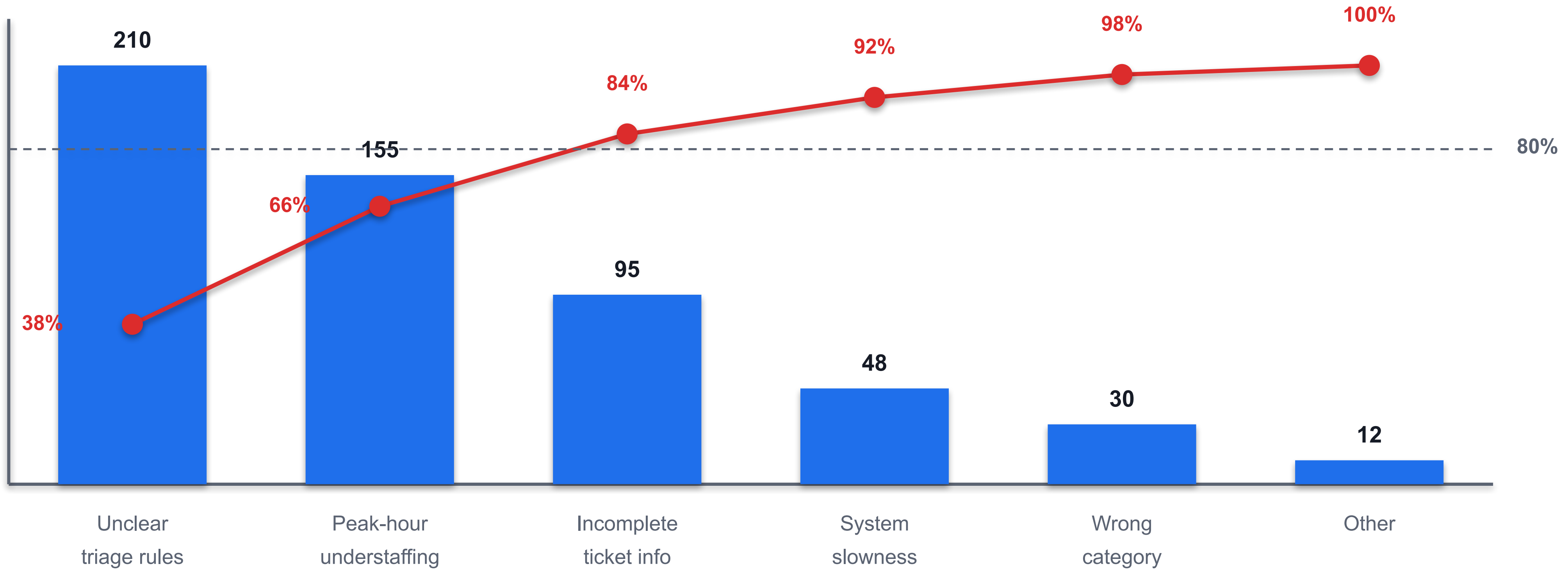
Treating common-cause variation as if it were special cause — called tampering — reliably makes the process worse, not better.

ANALYZE · PARETO

The Pareto Principle — the 80/20 rule

Roughly 80% of the effects come from 20% of the causes. Find that 20% and you get most of the improvement for a fraction of the effort.

Pareto Chart — Causes of Delayed Ticket Assignment



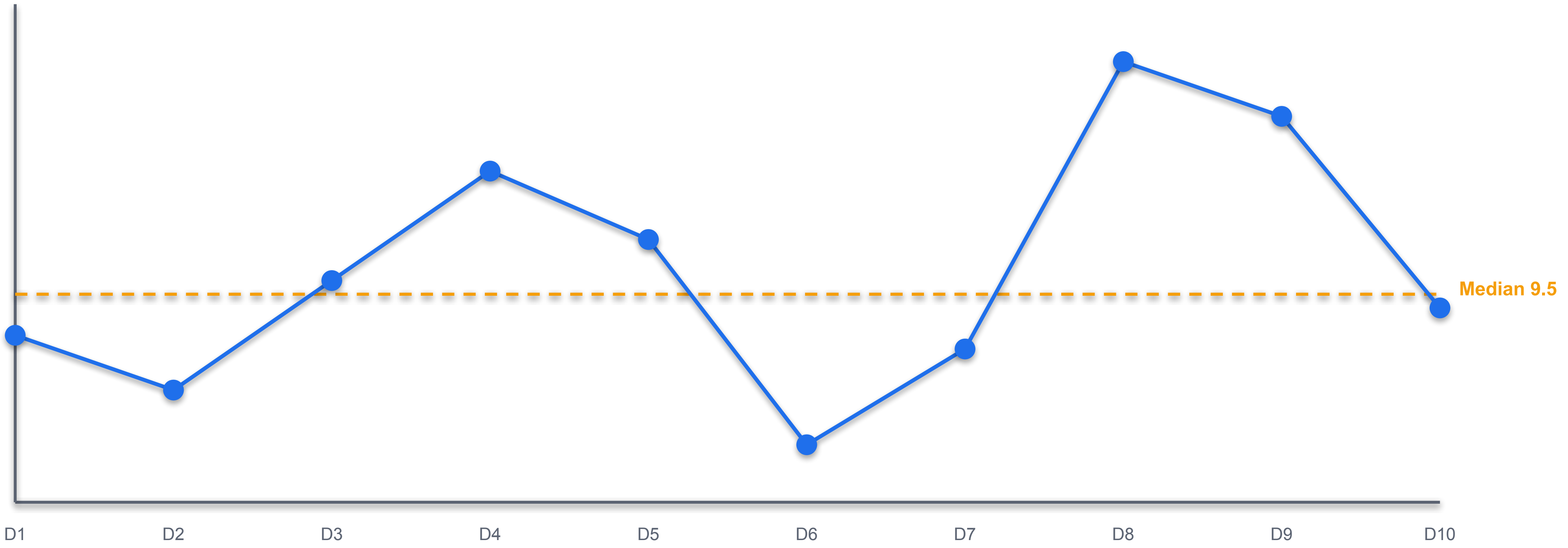
Two categories — unclear triage rules and peak-hour understaffing — account for about 66% of all delays. Start there.

PARETO · METHOD

How to Build a Pareto Chart

1. Agree the categories and make them mutually exclusive
2. Count the occurrences in each category
3. Sort the categories in descending order
4. Calculate the cumulative percentage
5. Plot bars plus the cumulative line and read off the vital few

Run Chart — Average Ticket Assignment Time by Day



A single point above the median is normal variation. A run of 7+ points on one side signals a real shift.

Reading a Run Chart — Five Patterns

1 **Trend**
7 or more consecutive points rising or falling — the process is drifting.

2 **Shift**
8 or more consecutive points on one side of the median — something changed.

3 **Cluster**
Points grouped together — suggests a periodic or batch effect.

4 **Oscillation**
Rapid up-down swings — the process is unstable, often from over-adjustment.

5 **Mixture**
Points avoiding the centre line — often two different processes combined.

6 **No pattern**
Random scatter around the median — stable, common-cause variation only.

ANALYZE · ROOT CAUSE

What is a root cause?

The deepest cause in the chain that you can actually act on — remove it, and the problem does not come back.

WHY EVIDENCE MATTERS

Fixing the wrong cause wastes the whole project.

If the analysis stops at the first plausible explanation, the team implements a change, the metric does not move, and confidence in the method is lost.

The 5 Whys Technique

1

Start with the problem

State the effect precisely, using the data you measured.

2

Ask why — and answer with evidence

Each answer must be something you can point to, not a guess.

3

Repeat about five times

Five is a guide, not a rule — stop when you reach an actionable cause.

4

Watch for blame

If an answer names a person rather than a process, ask why again.

5

Stop at actionable

The root cause must be something within the team's power to change.

6

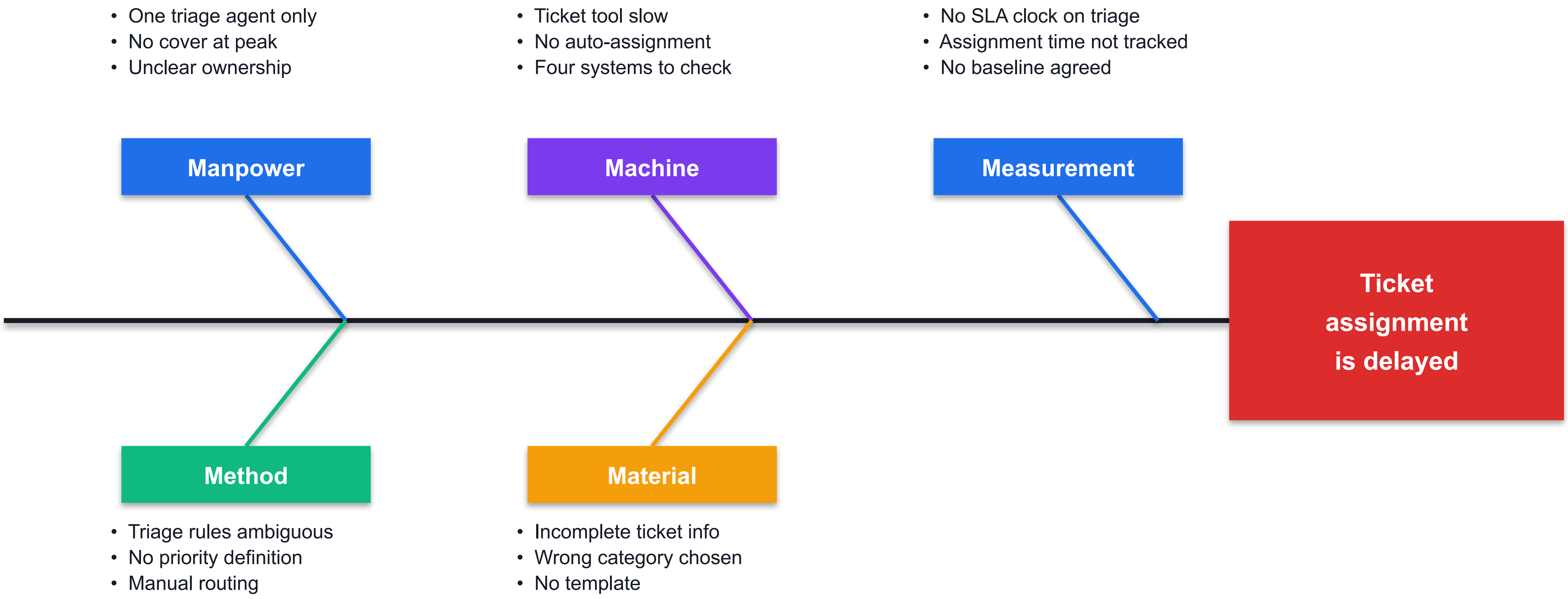
Verify with data

Test the final cause against the evidence before acting on it.

5 Whys — Worked Example: Ticket Assignment Delay

1. PROBLEM: Tickets take 9.4 hours to assign vs a 4-hour target
2. WHY? Tickets wait in the triage queue
3. WHY? Only one agent triages, and only in the morning
4. WHY? Triage is not in anyone's core role definition
5. ROOT CAUSE: Triage ownership was never assigned when the process changed

Fishbone (Ishikawa) — Ticket Assignment Is Delayed



Building a Fishbone Diagram

1. **Write the effect**
   Put the measured problem in the head of the fish — be specific.

2. **Choose the categories**
   5M: Manpower, Method, Machine, Material, Measurement. Service teams often use 4P.

3. **Brainstorm into each bone**
   Generate freely — no evaluation or debate during generation.

4. **Go three levels deep**
   Ask 'why does that happen?' on each cause to reach the real driver.

5. **Shortlist with multi-voting**
   Each team member votes; the causes with most votes go forward.

6. **Verify with data**
   A cause on the diagram is a HYPOTHESIS until the data supports it.

Brainstorming — Getting Good Causes Out

1

Quantity first

Generate a long list before evaluating anything.

2

No criticism

Judging ideas during generation stops people contributing.

3

Build on ideas

One person's half-idea is often another's breakthrough.

4

Everyone contributes

Use round-robin or brainwriting so the loudest voice does not dominate.

Multi-Voting — Narrowing the List

1

Why use it

A long cause list needs to become a short, agreed shortlist.

2

How it works

Each member gets N votes (about a third of the number of items).

3

Vote independently

Vote privately first so nobody anchors on the manager's choice.

4

Then verify

The top-voted causes still have to be tested against the data.

Interactive Problem-Solving Tools You Will Use

1

5 Whys

Build and share the 5 Whys chain online — alfredang.github.io/5whys/

2

Fishbone Diagram

Ishikawa cause-and-effect builder — alfredang.github.io/fishbone/

3

Pareto Chart (collaborative)

Your team brainstorms and votes in one live session; the Pareto chart builds itself — alfredang.github.io/paretochart/

4

NovaSPC

Run charts, SPC charts and process capability from your own CSV — alfredang.github.io/novaspc/

Using the Collaborative Pareto Tool in Your Team

1. Facilitator creates a session and shares the access code
2. Every team member joins with the code
3. The team brainstorms candidate causes into the session
4. Each member votes on the causes that matter most
5. The live Pareto chart reveals the vital few

WHAT YOU'LL DO

Hands-On Labs — Analyze

1

Lab 7 — Pareto, Run Charts, Variation, Yield, DPU, and

A Pareto chart, a run chart and calculated yield, DPU, DPO, DPMO and s

2

Lab 8 — Root Cause Analysis with 5 Whys, Fishbone, and

A completed 5 Whys chain, a Fishbone diagram and an evidence-tested ca

DMAIC · ANALYZE · HANDS-ON

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

LAB 7

Turn collected data into decisions. Apply the Pareto principle to find the vital few categories, use a run chart to see behaviour over time, and calculate the standard Six Sigma process metrics.

You'll build

A Pareto chart, a run chart and calculated yield, DPU, DPO, DPMO and sigma level.

Tools & techniques: Pareto Chart tool (collaborative), NovaSPC, yield/DPU/DPO/DPMO, sigma level, variation

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 1 OF 8

1

Build the Pareto table: category, count, percentage and cumulative percentage.

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 2 OF 8

2

In your group, open the collaborative Pareto tool — one member creates the session and shares the code, then everyone joins, brainstorms and votes to produce the live Pareto chart.

```
https://alfredang.github.io/paretochart/
```

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 3 OF 8

3

Identify the vital few categories driving about 80% of the problem.

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 4 OF 8

4

Export your assignment-time data as CSV and plot a run chart in NovaSPC.

```
https://alfredang.github.io/novaspc/
```

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 5 OF 8

5

Interpret the run chart for trend, shift, cluster and oscillation patterns.

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 6 OF 8

6

Calculate yield, DPU, DPO and DPMO from your defect data.

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 7 OF 8

7

Convert DPMO to a sigma level and interpret what it says about the process.

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 8 OF 8

8

Distinguish common-cause from special-cause variation and why the response differs.

LAB 7 · VERIFY

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

✅ Check your work

Your cumulative percentage column reaches 100%, and you can state the sigma level with the DPMO it came from.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

LAB 8

Move from symptom to cause. Apply the 5 Whys to drill past the obvious, organise candidate causes on a Fishbone diagram, then test each candidate against the evidence you actually collected.

You'll build

A completed 5 Whys chain, a Fishbone diagram and an evidence-tested cause shortlist.

Tools & techniques: 5 Whys tool, Fishbone tool, 5M categories, brainstorming, multi-voting

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 1 OF 7

1

Write the problem statement precisely — the effect you are explaining.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 2 OF 7

2

Complete a 5 Whys chain with the online 5 Whys tool, continuing until you reach an actionable cause.

```
https://alfredang.github.io/5whys/
```

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 3 OF 7

3

Build a Fishbone diagram with the online Fishbone tool using the 5M categories: Manpower, Method, Machine, Material, Measurement.

```
https://alfredang.github.io/fishbone/
```

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 4 OF 7

4

Brainstorm candidate causes into each category — no evaluation during generation.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 5 OF 7

5

Use multi-voting to shortlist the most likely causes as a team.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 6 OF 7

6

Test each shortlisted cause against your Lab 7 data — does the evidence support it?

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 7 OF 7

7

State why the team must not jump straight to solutions.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

✅ Check your work

Each shortlisted root cause is supported by named evidence, and your 5 Whys chain ends at something you can act on.

PHASE RECAP

Recap — Analyze

- Variation — Common cause is built into the process; special cause is an external, assignable signal.
- Pareto analysis — The 80/20 rule separates the vital few causes from the trivial many.
- Run charts — Plot the metric over time to reveal trend, shift, cluster and oscillation.
- 5 Whys — Ask why repeatedly to drill from the symptom down to an actionable cause.
- Fishbone diagram — Organise candidate causes by category — Manpower, Method, Machine, Material, Measurement.
- Evidence testing — A cause is only a root cause when the data you collected supports it.

DMAIC · IMPROVE

Improve — Fix the Cause

Countermeasures · 5S · Poka-Yoke · Standard work · Kaizen · Solution selection · Piloting

Key Concepts — Improve

1 **Generating solutions**
Brainstorm widely against the proven root cause before evaluating anything.

2 **Solution selection**
Score candidate solutions against weighted criteria before committing.

3 **5S**
Sort, Set in order, Shine, Standardise, Sustain — for physical and digital work.

4 **Poka-Yoke**
Mistake proofing makes the error difficult or impossible to make in the first place.

5 **Standard work**
Document the improved method so anyone can repeat it the same way.

6 **Piloting**
Test the change at small scale to expose issues before full rollout.

DMAIC · I — IMPROVE

Improve — What change will actually fix it?

The Improve phase generates, selects and pilots solutions that address the proven root cause.

Steps in the Improve Phase

1. Generate candidate solutions against the root cause
2. Select using weighted criteria
3. Pilot the change at small scale
4. Measure whether the metric actually moved
5. Standardise the new method

Generating Solutions

1

Brainstorming

Free generation against the specific root cause — not the general problem.

2

Brainwriting

Written idea generation; avoids the loudest voice dominating the room.

3

Benchmarking

Find who already does this well, internally or externally, and learn from them.

4

Anti-brainstorming

Ask how to make the problem worse, then invert every answer.

Prioritising Countermeasures — Impact vs Effort

IMPACT →	**Quick wins** High impact, low effort. Do these first — they build momentum and credibility.	**Major projects** High impact, high effort. Worth doing, but plan and resource them properly.
	Fill-ins Low impact, low effort. Do them if there is spare capacity.	**Thankless tasks** Low impact, high effort. Avoid these entirely.

EFFORT REQUIRED →

The Solution Selection Matrix

1

List the criteria

Effectiveness, cost, time to implement, risk, ease of adoption.

2

Weight each criterion

Not all criteria matter equally — agree the weights first.

3

Score each solution

Score every candidate against each criterion on a consistent scale.

4

Rank by weighted total

Multiply score by weight, sum, and rank — the decision becomes defensible.

5S — Organising the Workplace

1. SORT — remove what is not needed
2. SET IN ORDER — a place for everything
3. SHINE — clean and inspect regularly
4. STANDARDISE — make the first 3S the norm
5. SUSTAIN — audit and keep the discipline

5S Applied to Digital Work

1 **Sort**
Archive the ticket queues, folders and dashboards nobody uses.

2 **Set in order**
One agreed place for templates, runbooks and knowledge articles.

3 **Shine**
Regularly review and retire out-of-date knowledge base entries.

4 **Standardise**
Agreed naming conventions and ticket templates for everyone.

5 **Sustain**
A short monthly review to check the standard is still being followed.

6 **Why it works**
Less searching, fewer mistakes, faster onboarding for new agents.

IMPROVE · POKA-YOKE

Poka-Yoke — mistake proofing

Design the process so the error is difficult or impossible to make — rather than relying on people to be careful.

IMPROVE · POKA-YOKE LEVELS

Three Levels of Mistake Proofing

Prevention

The error cannot happen

- Strongest form — design it out.
- A required field blocks submission.
- A connector only fits one way.

Detection

The error is caught immediately

- The error happens but is flagged at once.
- Validation warns on an invalid category.
- Weaker, but often easier to add.

Mitigation

The impact is reduced

- The error happens and is not caught.
- But its consequence is limited.
- Weakest — use only when the others cannot apply.

Standard Work

1

What it is

The documented, current best-known way to perform the task.

2

Why it matters

You cannot improve a process that everyone performs differently.

3

What it contains

The sequence, the time each step takes and the quality checks.

4

Who writes it

The people who do the work — not a manager writing in isolation.

5

Keep it living

Update it every time a better method is proven.

6

The Lean insight

Standard work is the baseline for the NEXT improvement, not a straitjacket.

Kaizen — Continuous Small Improvement

1

The principle

Many small improvements beat waiting for one perfect large one.

2

Who does it

Everyone — Kaizen is explicitly not reserved for specialists.

3

Kaizen event

A focused 3-5 day workshop to fix one process end to end.

4

Yellow Belt fit

This is exactly the scale of improvement a Yellow Belt leads.

Piloting Before Full Rollout

1 **Why pilot**
Exposes practical issues cheaply, before they affect every customer.

2 **Keep it small**
One team, one shift or one ticket category is usually enough.

3 **Measure the same way**
Use the identical operational definition as your baseline, or you cannot compare.

4 **Run it long enough**
Cover a full business cycle so you see normal variation.

5 **Decide honestly**
Adopt, adapt or abandon — a failed pilot that saves a bad rollout is a success.

6 **Then standardise**
Only after the pilot proves the gain do you write it into standard work.

WHAT YOU'LL DO

Hands-On Labs — Improve

1. **Lab 9 — Countermeasures, 5S, Mistake Proofing, Standar**
   A prioritised countermeasure set with a 5S plan, a poka-yoke design an

2. **Lab 13 (elective) — Solution Selection Matrix, Benchmarking,**
   A scored solution selection matrix, a benchmarking summary and an FMEA

DMAIC · IMPROVE · HANDS-ON

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

LAB 9

Generate and select countermeasures that attack the root cause you proved — not the symptom. Apply the core Lean improvement tools: 5S, poka-yoke, visual management, standard work and Kaizen.

You'll build

A prioritised countermeasure set with a 5S plan, a poka-yoke design and standard work.

Tools & techniques: Brainstorming, 5S, poka-yoke, visual management, standard work, Kaizen

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 1 OF 7

1

Generate countermeasures for each proven root cause — quantity before quality.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 2 OF 7

2

Prioritise using an impact vs effort matrix to find the quick wins.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 3 OF 7

3

Apply 5S thinking — Sort, Set in order, Shine, Standardise, Sustain — including to digital work.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 4 OF 7

4

Design one poka-yoke that makes the error difficult or impossible to make.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 5 OF 7

5

Draft standard work so the improved method is repeatable by anyone.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 6 OF 7

6

Plan a Kaizen event or pilot to test the countermeasure at small scale first.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 7 OF 7

7

Distinguish a containment countermeasure from a permanent solution.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

✅ Check your work

Every countermeasure traces back to a proven root cause, and your poka-yoke prevents rather than detects the error.

DMAIC · IMPROVE · HANDS-ON

Elective — Solution Selection Matrix, Benchmarking, and FMEA

LAB 13

ELECTIVE

Three Improve-phase tools from the original course. Score solutions against weighted criteria, benchmark against outstanding practice, and use FMEA to find how a proposed change could fail before it does.

You'll build

A scored solution selection matrix, a benchmarking summary and an FMEA with RPN.

Tools & techniques: Solution selection matrix, benchmarking, FMEA, RPN scoring

Elective — Solution Selection Matrix, Benchmarking, and FMEA

STEP 1 OF 5

1

Agree the selection criteria and weight each by importance.

Elective — Solution Selection Matrix, Benchmarking, and FMEA

STEP 2 OF 5

2

Score every candidate solution against the weighted criteria and rank the results.

Elective — Solution Selection Matrix, Benchmarking, and FMEA

STEP 3 OF 5

3

Benchmark your process against an internal or external reference and note the gap.

Elective — Solution Selection Matrix, Benchmarking, and FMEA

STEP 4 OF 5

4

Build an FMEA: failure mode, effect, cause, then score Severity, Occurrence and Detection.

Elective — Solution Selection Matrix, Benchmarking, and FMEA

STEP 5 OF 5

5

Calculate RPN = S x O x D and address the highest-RPN failure modes first.

Elective — Solution Selection Matrix, Benchmarking, and FMEA

✅ Check your work

Your matrix ranks solutions by weighted score, and every FMEA row has an RPN and an action for the highest scores.

PHASE RECAP

Recap — Improve

- Generating solutions — Brainstorm widely against the proven root cause before evaluating anything.
- Solution selection — Score candidate solutions against weighted criteria before committing.
- 5S — Sort, Set in order, Shine, Standardise, Sustain — for physical and digital work.
- Poka-Yoke — Mistake proofing makes the error difficult or impossible to make in the first place.
- Standard work — Document the improved method so anyone can repeat it the same way.
- Piloting — Test the change at small scale to expose issues before full rollout.

C

DMAIC · CONTROL

Control — Hold the Gain

Control plan · Visual management · SOPs · Huddles · A3 · Handover · Certification

Key Concepts — Control

1 **Control plan**
Metric, target, monitoring method, frequency, owner and reaction plan.

2 **Process control**
A process is in control when it varies consistently within expected limits.

3 **Visual management**
Boards and dashboards keep the improved performance visible to everyone.

4 **Standard operating procedures**
Written instructions that lock in the improved method.

5 **Team huddles**
Short, regular stand-ups that surface problems while they are still small.

6 **A3 and handover**
One-page storytelling to summarise, hand over and sustain the improvement.

DMAIC · C — CONTROL

Control — How do we make the gain stick?

Without the Control phase, processes drift back to the old way within months and the whole project is wasted.

Steps in the Control Phase

1. Build the control plan
2. Set up monitoring and visual management
3. Document the SOP and standard work
4. Hand over to the process owner
5. Close the project and share the learning

The Control Plan — Six Required Columns

1. **Metric**
   What is being monitored — the same operational definition as the baseline.

2. **Target**
   The agreed acceptable value or range.

3. **Method**
   How the measurement is taken — report, dashboard, sample or audit.

4. **Frequency**
   How often it is checked: per shift, daily, weekly.

5. **Owner**
   The named person accountable — a role, never 'the team'.

6. **Reaction plan**
   Exactly what happens when the metric falls outside target.

CONTROL · ACCOUNTABILITY

A control plan without a named owner is a wish.

Every row needs a person accountable and a specific action to take when the number goes wrong — otherwise nothing happens.

CONTROL · PROCESS CONTROL

What is process control?

A process is 'in control' when it varies consistently within expected limits over time — predictable, even if not yet perfect.

Control Limits vs Specification Limits

Set by the customer

- Specification limits (LSL / USL)
  - Given by the CUSTOMER
  - Define what counts as a defect
  - Do not change when the process changes
  - Outside them = a defect

Set by the process

- Control limits (LCL / UCL)
  - Calculated from the PROCESS itself
  - Typically the mean ± 3 standard deviations
  - Change when the process changes
  - Outside them = a special cause to investigate

Statistical Process Control (SPC)

1

What it is

Monitoring a process over time so problems are predicted, not discovered late.

2

The core idea

Plot the metric against control limits and watch for non-random patterns.

3

Why it saves money

Preventing defects always costs less than inspecting and reworking them.

4

Yellow Belt level

You are expected to read and interpret a control chart, not to construct one.

Visual Management

1. **Visual boards**
Display the metric, the target and the current status where the team works.

2. **Make it obvious**
Anyone should see whether performance is on target within a few seconds.

3. **Team huddles**
A short daily stand-up at the board — 15 minutes maximum.

4. **Drives ownership**
When performance is visible, the team owns it rather than the manager.

5. **Surfaces issues early**
Problems are raised while they are still small and cheap to fix.

6. **Keep it current**
An out-of-date board is worse than no board — it teaches people to ignore it.

Standard Operating Procedures (SOPs)

1

Written instructions

Describe how to perform the task to achieve the required result.

2

Include the measures

State the benchmarks and quality checks, not only the steps.

3

Centrally accessible

Held in one agreed place everyone can reach and search.

4

Version controlled

It must be obvious which version is current.

Go to the Gemba

1 **Gemba means 'the real place'**
Where the work actually happens — not the meeting room.

2 **Go and see**
Observe the process directly rather than relying on reports about it.

3 **Ask, do not blame**
The purpose is to understand the process, never to audit the person.

4 **Why it works**
Reports show what was recorded; Gemba shows what actually happens.

CONTROL · THE A3 REPORT

The A3 — One Page That Tells the Whole Story

1 **Background**
Why this problem was worth working on.

2 **Current state**
The baseline, with the data and process map.

3 **Goal**
The target condition, stated as a measurable number.

4 **Root cause analysis**
What the 5 Whys and Fishbone actually proved.

5 **Countermeasures**
What was changed, and why that addresses the proven cause.

6 **Results & follow-up**
Whether the metric moved, and who now owns the control plan.

Project Handover and Closure

1. Confirm the improvement held for a full cycle
2. Hand the control plan to the named process owner
3. Train everyone on the new standard work
4. Document the financial or service benefit
5. Share the learning so others can reuse it

WHAT YOU'LL DO

Hands-On Labs — Control

1

Lab 10 — Control Plan, A3 Summary, Handover, and Certif

A control plan, an A3 one-page summary, a handover checklist and a rea

2

Lab 14 (elective) — Descriptive Statistics and Implementatio

A descriptive statistics summary and a dated implementation plan.

DMAIC · CONTROL · HANDS-ON

Control Plan, A3 Summary, Handover, and Certification Readiness

LAB 10

Hold the gain. Build a control plan that names the metric, the target, the monitoring frequency, the owner and the reaction plan — then tell the whole improvement story on a single A3 page and hand it over.

You'll build

A control plan, an A3 one-page summary, a handover checklist and a readiness plan.

Tools & techniques: Control plan, SPC thinking, visual management, A3 report, handover

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 1 OF 6

1

Create the control plan: metric, target, monitoring method, frequency, owner, reaction plan.

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 2 OF 6

2

Define the visual management approach — board, dashboard or huddle — that keeps it visible.

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 3 OF 6

3

Specify the reaction plan: exactly what happens when a metric falls outside target.

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 4 OF 6

4

Write the A3 summary: background, current state, goal, analysis, countermeasures, results, follow-up.

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 5 OF 6

5

Prepare the handover checklist so the process owner can sustain it without you.

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 6 OF 6

6

Complete your personal certification readiness plan — which topics need most review.

Control Plan, A3 Summary, Handover, and Certification Readiness

✅ Check your work

Every control plan row has a named owner and a reaction plan, and your A3 fits on one page.

DMAIC · CONTROL · HANDS-ON

Elective — Descriptive Statistics and Implementation Planning

LAB 14

ELECTIVE

Close the loop. Compute the measures of central tendency and dispersion that describe your data, then convert the selected countermeasures into a dated implementation plan with owners and barriers identified.

You'll build

A descriptive statistics summary and a dated implementation plan.

Tools & techniques: Mean, median, range, standard deviation, implementation planning

Elective — Descriptive Statistics and Implementation Planning

STEP 1 OF 4

1

Compute the mean and median for your assignment-time data and compare them.

Elective — Descriptive Statistics and Implementation Planning

STEP 2 OF 4

2

Compute the range and standard deviation to describe the spread.

Elective — Descriptive Statistics and Implementation Planning

STEP 3 OF 4

3

Explain what the mean-versus-median difference reveals about outliers and skew.

Elective — Descriptive Statistics and Implementation Planning

STEP 4 OF 4

4

Write the implementation plan: action, owner, date, barriers and mitigation.

LAB 14 · VERIFY

Elective — Descriptive Statistics and Implementation Planning

✅ Check your work

You can explain why the mean and median differ in your data, and every implementation action has an owner and a date.

PHASE RECAP

Recap — Control

- Control plan — Metric, target, monitoring method, frequency, owner and reaction plan.
- Process control — A process is in control when it varies consistently within expected limits.
- Visual management — Boards and dashboards keep the improved performance visible to everyone.
- Standard operating procedures — Written instructions that lock in the improved method.
- Team huddles — Short, regular stand-ups that surface problems while they are still small.
- A3 and handover — One-page storytelling to summarise, hand over and sustain the improvement.

WRAP-UP

Course Summary & Next Steps

What You Achieved — The Full DMAIC Journey

D | M | A | I | C

Define

- Captured VOC and CTQ
- Wrote the charter
- Built the SIPOC
- Mapped the process

Measure

- Found the 8 wastes
- Built the data plan
- Collected the baseline
- Calculated DPMO

Analyze

- Built the Pareto
- Read the run chart
- Ran 5 Whys and Fishbone
- Proved the cause

Improve

- Selected solutions
- Applied 5S and poka-yoke
- Wrote standard work
- Piloted the change

Control

- Built the control plan
- Set visual management
- Wrote the A3
- Handed over

WHAT YOU BUILT

Your Integrated Improvement Package

1 **Role and scenario**
Yellow Belt responsibility table and a selected improvement scenario.

2 **VOC, CTQ and waste**
Customer requirements translated to CTQs, plus a waste walk log.

3 **SIPOC and process map**
Macro and detailed maps with pain points and handoffs marked.

4 **Charter and Define pack**
PDCA charter, problem statement, scope and stakeholder map.

5 **Measurement and analysis**
Data collection plan, check sheet, Pareto, run chart and metrics.

6 **Root cause and countermeasures**
5 Whys, Fishbone, evidence tests and prioritised countermeasures.

7 **Control plan and A3**
Control plan, visual management approach, A3 summary and handover.

8 **Certification readiness**
Your personal review plan for the assessment.

BEFORE THE ASSESSMENT

Final Readiness Checklist

1. Can you explain the Yellow Belt role and where it stops?
2. Can you define Lean, Six Sigma and Lean Six Sigma in your own words?
3. Can you trace a VOC statement through to a CTQ and a metric?
4. Can you name the eight wastes and give a service example of each?
5. Can you explain each DMAIC phase and what it delivers?
6. Can you calculate yield, DPU, DPO and DPMO from raw data?
7. Can you interpret a Pareto chart and a run chart?
8. Can you explain your root cause with the evidence behind it?
9. Can you describe how the control plan sustains the gain?

NEXT STEPS

Continuing Your Lean Six Sigma Journey

1 **Apply it at work**
Run one small PDCA improvement in your own area within 30 days.

2 **Green Belt**
The next step — leads smaller DMAIC projects and handles the statistics.

3 **Keep the templates**
Your lab outputs are reusable templates for real projects.

4 **Join the community**
Share improvements with colleagues; Kaizen spreads by example.

ASSESSMENT

Final Assessment

Written Assessment (SAQ, 60 minutes) followed by the Case Study (90 minutes). Both are open book.

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (SAQ), then the Case Study — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

Digital Attendance (Assessment)

1. Trainer displays the SSG digital attendance QR code
2. Scan the QR code with your phone camera
3. Key in your NRIC/FIN and submit
4. Attendance must be recorded before you begin the papers

YOUR FEEDBACK

TRAQOM Survey

1. Open the TRAQOM survey link on the LMS
2. Key in the last four characters of your NRIC/FIN
3. Key in the six-digit course run ID
4. Complete and submit — your feedback shapes this course

Certificate & Support

- Two e-certificates are awarded on demonstrating competency and achieving at least 75% attendance.
- A SkillsFuture Statement of Attainment (SOA) is issued for the WSQ assessment.
- Email: enquiry@tertiaryinfotech.com
- Tel / WhatsApp: +65 6100 0613

END OF COURSE

Thank You!

Go and improve one process this month — that is what a Yellow Belt is for.